PRACTICAL EXAM DESCRIPTION – PAGE 01

(Đề thi thực hành trên EOS)

I Case Study

QuickMart Vietnam (QVM) is a rapidly expanding retail business running a chain of over 200 convenient stores across major cities in Vietnam. To stay competitive and streamline multi-store operations, QVM's board of directors (BOD) has decided to develop a centralized **Convenient Store Chain Management System (CSCMS)** that replaces the company's fragmented, store-level spreadsheets and manual processes with a unified digital platform.

The CSCMS links store managers, warehouse staff, delivery drivers, and headquarters (HQ) administrators through a single platform. When a customer checks out, the cashier scans products at the point-of-sale (POS) terminal; the system records the transaction, applies any active promotions or loyalty discounts, and issues a printed or digital receipt. Store inventory is decremented in real time, and when any product falls below the re-order threshold, an automatic replenishment request is sent to the nearest warehouse. Warehouse staff receive pick-and-pack orders on their handheld devices, update the dispatch status after delivery, and the system reconciles stock across all locations automatically. Store managers can view shift schedules, log daily cash reconciliation, and track individual staff performance through a dashboard; any pay discrepancy triggers an alert to both the store manager and the HQ finance team. HQ administrators monitor chain-wide sales, margin, and inventory KPIs on a real-time analytics dashboard, configure product catalogues and pricing rules centrally, and push promotional campaigns to selected stores or all locations simultaneously. Suppliers are onboarded into the system and can submit electronic delivery notes that are matched against purchase orders, with discrepancies flagged for review before goods are accepted into stock.

On the security front, all users authenticate with role-based credentials, and multi-factor authentication (MFA) is enforced for manager and HQ accounts; all data in transit is encrypted with TLS 1.3, and data at rest is encrypted using AES-256. The system also keeps a tamper-evident audit log of every transaction, stock adjustment, and configuration change for at least three years to satisfy regulatory requirements. For scalability, the platform must support at least 200 concurrent POS sessions without performance degradation; the backend must scale horizontally through container orchestration so that adding new stores requires no manual infrastructure changes; and all inter-service APIs must be stateless and versioned to allow independent deployment of individual modules as the store network grows. Every POS interaction must respond within 1 second under peak load to avoid checkout queues; the web dashboard must load within 2 seconds; and the system must maintain 99.9% uptime with automated failover so that a regional power outage does not halt store operations.

The CSCMS replaces several disconnected legacy tools; each store currently operates independently, so integration requirements are expected to emerge and change incrementally as stores are on-boarded one by one. Development will be led by QVM's IT department together with a cross-functional team of **4–5 developers**, a business analyst, and store-operations representatives who provide continuous feedback. Leadership expects the first stores to go live within **2 months** and the full chain to be covered within **12 months**, with new features delivered and deployed continuously throughout.

II Questions

Assume that you have been assigned to implement this project. Read the description carefully, then answer the following questions:

1. Your development team is highly skilled and experienced. Your manager proposes applying the **Kanban model** to this project. Do you agree or disagree? List and explain all the main principles, practices, and characteristics of the **Kanban model**, then match each of them to the project characteristics above to justify your answer. **(2.0 points)**

2. Your manager wants the system tested from the smallest unit level up to the fully integrated system level. At the unit level, testing must focus on program structure; at higher levels, testing must be based on functional requirements.

- **a.** Which type and level/stage of testing does your manager require? **(0.7 point)**
- **b.** Who carries out testing at each stage/level, and why are they the best fit for that role? **(0.3 point)**

3. Based on the project description above, identify and list: **5 functional requirements, 3 security requirements**, and **2 scalability requirements**. **(2.0 points)**

4. Write **5 user stories** based on your answers in Question 3. **(1.5 points)**

5. The manager asks you to apply design patterns to the software system. List and describe in detail two design patterns discussed in the courses **(1.0 points)**.

6. Build a **story map** for the **"Manage Product Sales"** activities described in the project above. **(2.5 points)**

III Notes

a) Mỗi câu trả lời sai sẽ bị trừ 0.1 điểm b) Sau đây là hướng dẫn cách trình bày Story Map trên phần mềm EOS.

Cấu trúc tổ chức khi trình bày Story Map được chia làm 2 phần. Phần 1 **(A)** mô tả các activities và user tasks. Phần 2 **(B)** mô tả các user story cần thực hiện trong mỗi lần bàn giao. Các phần được trình bày như sau:

A. Activities and User tasks

1. <Actitivity 1 name>
    - 1.1 <User task 1 of activity 1>
    - 2.2 <User task 2 of activity 1>
2. <Actitivity 2 name>
    - 2.1 <User task 1 of activity 2>
    - 2.2 <User task 2 of activity 2>

....

B. Releases

Release 1---

<Liệt kê các user story cho từng user task thực hiện trong lần bàn giao đầu tiên. Chỉ mục của các user story tương ứng với các user task ở phần trên>

- 1.1.1 User story 1 of user task 1 in activity 1
- 1.1.2 User story 2 of user task 1 in activity 1
- 2.1.1 User story 1 of user task 1 in activity 2
- 2.1.2 User story 2 of user task 1 in activity 2

...

Release #---

<Liệt kê các user story cho từng user task thực hiện trong lần bàn giao #>

- 1.1.3 User story 3 of user task 1 in activity 1
- 2.1.3 User story 3 of user task 2 in activity 2

....

Ví dụ: Khi mô tả trên EOS một Story Map có cấu trúc biểu đồ như sau

[Hình ảnh: Sơ đồ Story Map mẫu (Steve Rogalsky, winnipegagilist.blogspot.com) gồm 4 activities (Organize Email, Manage Email, Manage Calendar, Manage Contacts) ở hàng trên cùng; mỗi activity có nhiều user tasks bên dưới được phân theo 3 Releases.]

Story Map này được mô tả trong EOS như dưới đây.

A. Activities and User Tasks

1. Organize email
   - 1.1 Search email
   - 1.2 File email
2. Manage email
   - 2.1 Compose email
   - 2.2 Read email
   - 2.3 Delete email
3. Manage Calendar
   - 3.1 View Calendar
   - 3.2 Create Appt
   - 3.3 Update Appt
   - 3.4 View Appt
4. Manage contacts

......

B. Releases

B. Releases

Release 1---

- 1.1.1 Search email by key word
- 1.2.1 Move emails
- 1.2.2 Create subfolders
- 2.1.1 Create and send basic email
- 2.1.2 Send RTF email
- 2.2.1 Open basic email
- 2.2.2 Open RTF email

....

Release 2 ---

- 1.1.2 Limit search to one field
- 1.1.3 Limit search to +one field
- 2.1.3 Send HTML email
- 2.1.4 Set email priority

....